

Raja Hammad Naseer

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Professional Summary

AI & ML Engineer with 2+ years of production experience: LLM fine-tuning (QLoRA/PEFT), RAG pipelines, agentic workflows, and computer vision systems deployed at scale. Recent hands-on work fine-tuning a 1.1B parameter LLM on a 4 GB GPU. Strong Python, PyTorch, and full ML lifecycle ownership from data to quantised deployment.

Technical Skills

LLM & Fine-Tuning	PyTorch, PEFT (QLoRA / LoRA), bitsandbytes, TRL, Hugging Face, Quantisation (GGUF, q8_0), llama.cpp, Ollama
LLM Applications	LangChain, OpenAI API, RAG Pipelines, Agentic Workflows, MCP, Prompt Engineering, ChromaDB, Vector Search
ML & Computer Vision	YOLOv8, ONNX Runtime, Roboflow, Pose Estimation, ByteTrack, Kalman Filter, Model Training & Fine-Tuning
Languages	Python (<i>primary</i>), JavaScript, TypeScript, SQL, C/C++ (Arduino)
Backend & MLOps	FastAPI, Node.js, REST API Design, Docker, CI/CD, Git, Linux
Data	NumPy, Pandas, SciPy, Matplotlib, MongoDB, MySQL, PostgreSQL

Work Experience

AI Engineer & Full-Stack Developer

2024 – 2025

Jataq Technologies — Pakistan

- Architected and deployed an end-to-end LLM-powered customer support system into a live production product: RAG pipeline design, prompt chain engineering, and REST API integration into existing web infrastructure.
- Shaped the company's AI tooling strategy through systematic benchmarking of competing LLM frameworks against real-world performance data — recommendations adopted directly into the production stack.
- Shipped multiple production web applications with zero reported downtime via Node.js/React.js systems and Docker-based CI/CD pipelines.

Artificial Intelligence Engineer

2023 – 2024

VisionTech360 — Pakistan

- Owned the complete ML pipeline for EMACS, a multi-camera face recognition access control platform deployed across multiple live camera installations: Roboflow labeling, custom YOLOv8 training, hyperparameter tuning, ONNX inference optimisation, and React.js real-time dashboard.
- Improved inference reliability under varying lighting conditions through augmentation strategies and architecture comparison; containerised all CV inference pipelines with Docker for repeatable production deployment.

Projects

MediChat-AI — LLM Fine-Tuning on Consumer Hardware

TinyLlama 1.1B, PyTorch, PEFT (QLoRA), bitsandbytes,

llama.cpp, ChromaDB, FastAPI

- Fine-tuned TinyLlama 1.1B entirely on a 4GB consumer GPU using 4-bit quantisation + LoRA adapters — only 1.13% of weights trainable, full training run fit in ~3GB VRAM.
- Built Python pipeline restructuring 28k WebMD Q&A pairs into structured 4-section medical responses; iterated dataset design after first run revealed template memorisation despite 87% token accuracy — traded surface metrics for real condition-specific output.
- Quantised merged adapter to GGUF (q8_0, 1.17GB), deployed via Ollama, layered ChromaDB RAG (19,725 embedded Q&As) and FastAPI backend. [Model on HF](#).

Drone Surveillance — Multi-Object Tracking Research

Python, YOLOv11, ByteTrack, Norfair, Kalman Filter, Microsoft AirSim

- TU Ilmenau Research Skills seminar paper. Built real-time MOT pipeline; benchmarked ByteTrack vs. Norfair quantitatively on MOTA, ID-switch rate, and FPS across simulated aerial flight maneuvers — produced concrete trade-off findings for resource-constrained UAV deployment.

AI Debate Agent — Tool-Augmented Reasoning

Python, LangChain, Gemini API, Web Search API

- Production-quality LLM agent with dynamic tool use: live web search grounding, chain-of-thought reasoning, structured argument/counterargument output with citation tracking.

TEC Thermal Control — System Identification

Python, NumPy, SciPy, Arduino, DS18B20

- TU Ilmenau control engineering research seminar. Built closed-loop PID system; logged 7000+ sensor samples, fit second-order PT2 model with `scipy.optimize.curve_fit` ($R^2=0.9943$, RMSE=0.12 °C); validated 24 °C → 10 °C setpoint with active disturbance rejection.

Education

M.Sc. Computer & Systems Engineering — Technische Universität Ilmenau, Germany

Oct 2025 – Present

B.Sc. Computer Science — SZABIST, Pakistan

2019 – 2023

Certifications & Languages

Certifications Node.js — The Complete Guide · Google Data Analytics Professional Certificate

Languages Urdu (Native) · English (C1 — full professional) · German (B1)